AML-3406

AL and ML Capstone Project

Capstone Project Weekly Progress Report

Semester	Fall-2024 Semester 3rd
Course Code	AML-3406
Section	Section 1
Project Title	Car Damage Detection
Group Name	Group B
Student names/Student IDs	Sakshi (C0908000) Bansil Patel (C0912873) – TL Harsh Mohile (C0912872) Meet Patel (C0910378) Rachit Bhatt (C0902810)
Reporting Week	Week 5
Faculty Supervisor	William Pourmajidi

1. Grade the level of collaboration from different aspects between team members: (use: good, medium, below expectation)

Sakshi	Bansil Patel	Harsh Mohile	Meet Patel	Rachit Bhatt
Good	Good	Good	Good	Good

2. Tasks outlined in the previous Weekly Progress Report for this reporting week at the individual level and group level

2.1 Individual tasks (each team member)

Sakshi: Research on CI/CD Pipelining for ML-Ops.
Bansil Patel & Meet Patel: Applying various models initially on the dataset.
Harsh Mohile & Rachit Bhatt: Research on Docker for Deployment.

2.2 Your team's tasks

- Exploring and understanding the dataset.
- Research on Docker for Deployment.

3. Progress made in Reporting Week at an individual level and group level

3.1 Individual progress

Sakshi: Studied the theoretical concepts of CI/CD.
Bansil Patel: Applied regular NN with Model Evaluation and attempted in accessing the annotated data in the JSON files.
Harsh Mohile & Rachit Bhatt: Collaborated to create a demo project using Docker.

3.2 Your team's progress

- Worked on understanding Docker by creating a demo project; GitHub - Docker.

4. The areas/tasks you could not make progress and/or complete as scheduled or the difficulties encountered in this reporting week at individual level and group level.

4.1 Individual project blockers

Sakshi: N/A
Bansil Patel & Meet Patel: AttributeError exception generated when trying to access the JSON files.
Harsh Mohile & Rachit Bhatt: Docker installation and implementation on Windows platform was a little challenging when finding proper content for research and development.

4.2 Your team's project blockers

- Faced a progress blocker in the form of exception when trailing with JSON files; seems like an OS-level error due to some path or incorrect data provided to the library.

5. Tasks to be completed in next week at individual level and group level

5.1 Individual tasks

Sakshi: Implementation CI/CD on a demo project.

Bansil Patel | Harsh Mohile | Meet Patel: Resolve the exception and implement multi-model approach.

Rachit Bhatt: Start Implementing CI/CD using Docker through GitHub.

5.2 Your team's tasks

- Work on multi-model setup.
- Implement CI/CD using Docker with GitHub.

6. Include the tasks from your sprint planning (Github/Zenhub) for the present period.

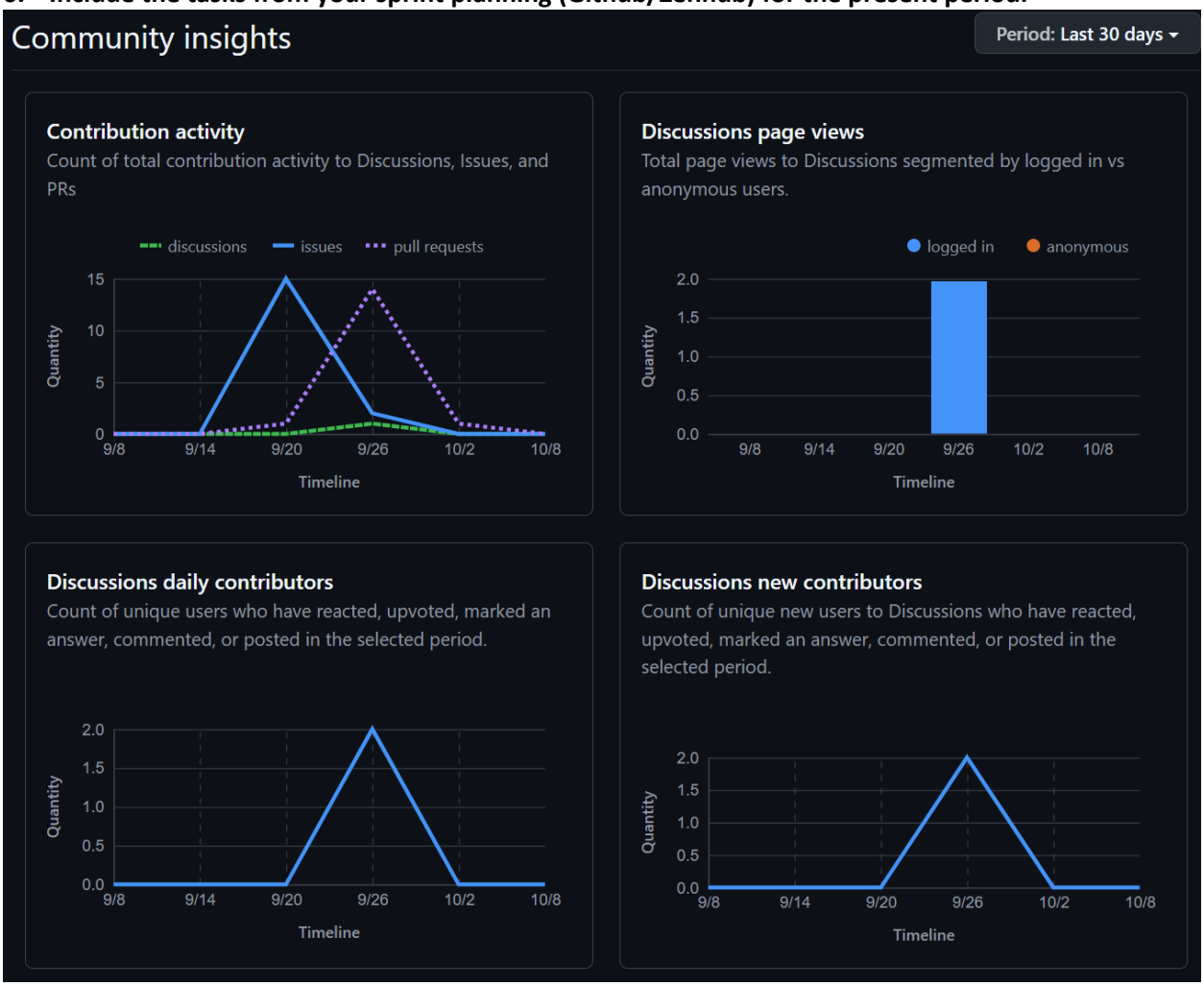

Figure 1: Weekly Activities

- In this week, there were 15 issues resolved with numerous PRs, contributing in development.

7. Include charts/graphs (e.g., burn down charts) from your project management tool (Github/Zenhub) that shows your progress for the period of this report.

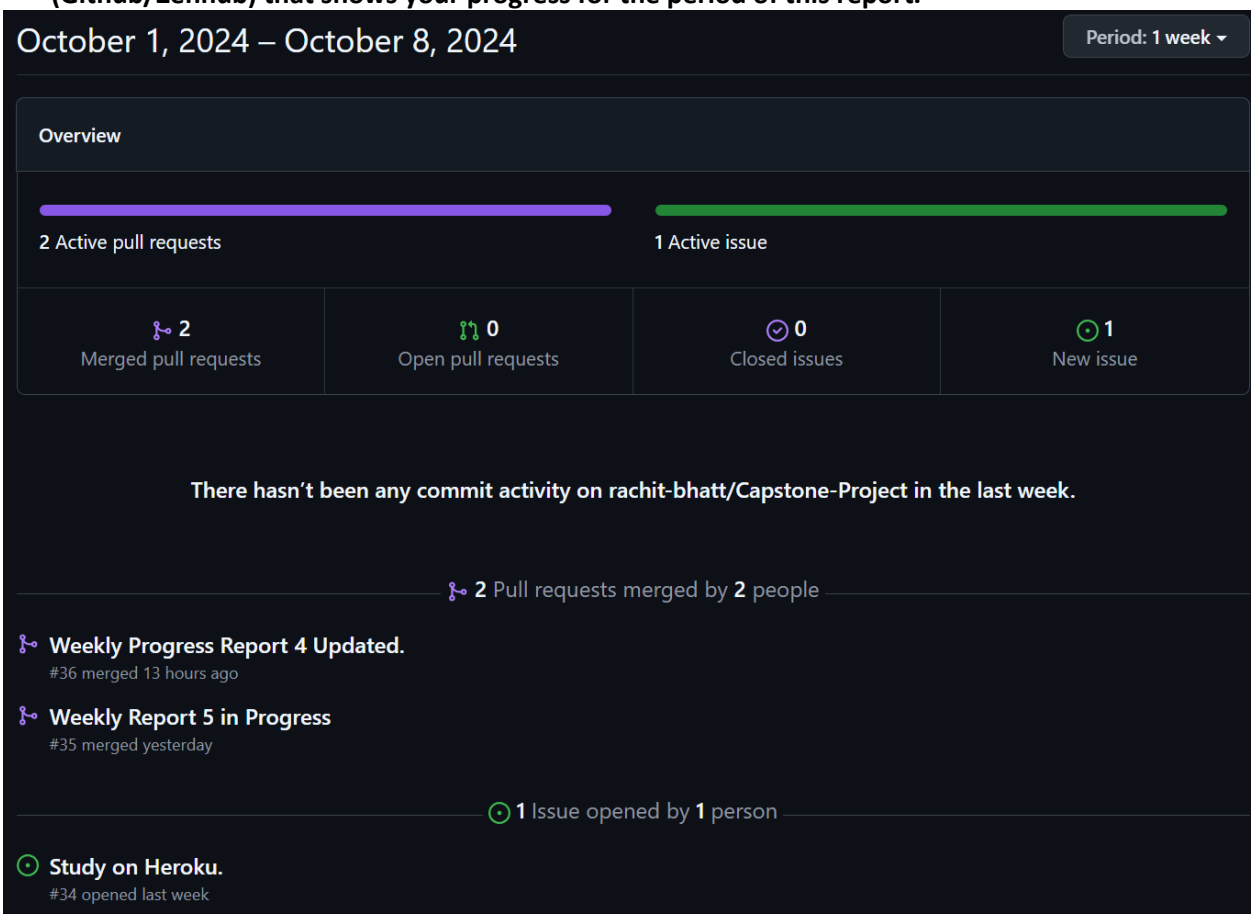

Figure 2: Contributors in the Week

- This week, there were two pull requests generated and merged that included phase 1 of our project and documentation.
- The one active issue we dealt with was the exception generated as shown in the *Figure 3*, took a lot of time and decided to shift it to the next scrum.

8. Include a note and address to your project Github with list of codes uploaded/updated on Github in this reporting week.

GitHub: https://www.github.com/rachit-bhatt/Capstone-Project

```
            predict_and_display_image('uploaded_image.jpg', model_eff)

# Attach the function to the upload button
upload_btn.observe(on_upload_change, names='value')

# Display the upload button and output widget
display(upload_btn, output)

```

```
Could not render content for 'application/vnd.jupyter.widget-view+json'
{"model_id":"428ebbf54e264659a871ca536e14de18","version_major":2,"version_minor":0}

Could not render content for 'application/vnd.jupyter.widget-view+json'
{"model_id":"4f7b00a88f9c4deb8036a8a75e08fa71","version_major":2,"version_minor":0}

---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
File c:\Users\pbans\OneDrive\Desktop\CD2\.conda\Lib\site-packages\ipywidgets\widgets\widget.py:773, in Widget._handle_msg(self, msg)
    771         if 'buffer_paths' in data:
    772             _put_buffers(state, data['buffer_paths'], msg['buffers'])
--> 773         self.set_state(state)
    775 # Handle a state request.
    776 elif method == 'request_state':

File c:\Users\pbans\OneDrive\Desktop\CD2\.conda\Lib\site-packages\ipywidgets\widgets\widget.py:650, in Widget.set_state(self, sync_data)
    645         self._send(msg, buffers=echo_buffers)
    647 # The order of these context managers is important. Properties must
    648 # be locked when the hold_trait_notification context manager is
    649 # released and notifications are fired.
--> 650 with self._lock_property(**sync_data), self.hold_trait_notifications():
    651     for name in sync_data:
    652         if name in self.keys:

File c:\Users\pbans\OneDrive\Desktop\CD2\.conda\Lib\contextlib.py:144, in _GeneratorContextManager.__exit__(self, typ, value, traceback)
    142 if typ is None:
    143     try:
--> 144         next(self.gen)
    145     except StopIteration:
    146         return False
...
     41         # Save the uploaded image to a temporary path
     42         with open('uploaded_image.jpg', 'wb') as f:
     43             f.write(file_info['content'])

AttributeError: 'tuple' object has no attribute 'items'
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...
```

Figure 3: Exception – Progress Blocker

- The exception when dealing with the image by the end user.
- Expecting it to be resolved in the next scrum.